SIMULTANEOUS GO VIA QUANTUM COLLAPSE

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Abstract. We construct a symmetric, simultaneous, deterministic evolution game *SGo*, which is in a certain mathematical sense a symmetrization of the classical board game Go. *SGo* is in some ways a simpler game than Go, as Komi, Ko and suicide rules are removed. On the other hand it has similar dynamics and move sensitivity, enabled by certain deterministic "quantum state" reduction, so that state evolution is deterministic. Using the argument of Nash, we show that *SGo* has a mixed equilibrium strategy to draw on average. It is an open problem if it has a strategy to draw (i.e. a pure Nash equilibrium drawing strategy).

1. Introduction

The game Go is an ancient game of local to global geometric principles. Like all sequential games it suffers from time asymmetry between the players Black and White. In Go this is addressed by an ad hoc rule called Komi: whereby the player White which moves second is given a specific point advantage. For Go, time asymmetry also results in certain pathologies in game states, for example *Ko* situation, where after a capture of a stone the opposing player may immediately recapture, but is forbidden by an extra rule. The latter sometimes leads to counterintuitive behavior, as the game loses locality.

At the same time, precisely because of its local to global nature Go is "almost time symmetric". We make this precise by constructing a symmetric, simultaneous (stateful) game *SGo* that is in one mathematical sense a symmetrization of Go, and on the other hand has very similar game dynamics to *Go*. *SGo* is a deterministic game, which is playable on a standard Go board, although we need a third type of stones that we call red. It is in some ways a simpler game than Go as it loses the Komi, Ko and suicide rules, and it has just two rules stone placement and capture (aside from end of game rule), but the latter rules and game states are now a bit more complicated, as we have to deterministically resolve simultaneous action.

The idea for *SGo* is loosely inspired by quantum mechanics. In a sense we branch the game state whenever the moves of the two players do not commute in classical *Go*. However, by itself this leads to a game with uninteresting dynamics. [1] To get interesting dynamics, we also have "objective reduction of the quantum state", it is a kind of collapse of the wave function arising deterministically. The term objective reduction is coined by Penrose, see for instance [2].

Using the famous idea of Nash in [1], we show that *SGo* has a mixed equilibrium strategy to draw on average. That said finding this even approximately is likely only possible on very small boards, due to the unreal size of the strategy space. We can also ask if *SGo* has a strategy to draw not just draw on average.

[1] It is just an example of the "universal symmetrization" of a sequential game.

2. Game rules

The game *SGo* is initialized with an empty standard Go board, with horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in Go, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in *SGo* they do this on the same turn without knowledge of the other's move on this turn. This requires modifying the placement and capture rules of Go, while removing the Komi rule, Ko, and suicide rules. We also need to introduce a third type of stones called red, which may be thought of as simultaneously white and black.

2.1. **Some definitions.** Given a board state S, that is any (at the moment no legality restrictions) collection of white, black, red stones on the board, let $G(S)$ denote the graph with vertices stones, and an edge between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency, always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A ***unicolor component*** C is a connected component of the full (a.k.a. induced) subgraph of $G(S)$ whose vertices are either all the black, or all the white stones. (Full subgraph means that all the edges of the original graph, between any pair of vertices of the subgraph, are included.) In this case we call C ***black***, respectively ***white***.

As in classical Go, a connected component C ***has no liberties*** if it cannot be extended (as a connected component) by placing a stone on the board. We will also call C a ***trapped component***.

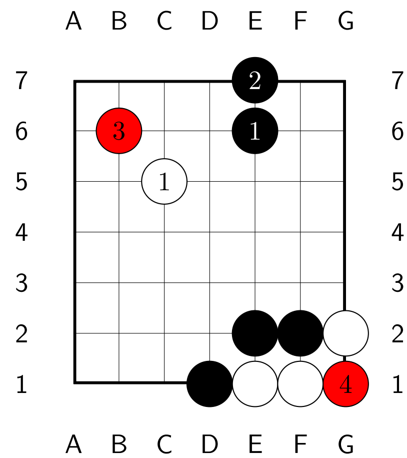
We define stones ***adjacent*** to C, as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

Given a unicolor component C, a white/black stone on the board is called a *trapping* if it is adjacent to a trapped unicolor component of opposite color.

2.2. **Stone placement resolution rule of *SGo*.** If the players move to the same intersection i on the board we place a stone there whose color resolves in the following decreasing order of priority:

(1) If either a white or black stone placed at i is trapping, then a red stone is placed at i. We call this a *trapping move*.
(2) If the intersection i is adjacent to a black stone, and neither a white stone nor red stone, a black stone is placed. If the intersection is adjacent to a white stone and neither a black stone nor red stone, a white stone is placed.
(3) Otherwise, a red stone is placed.

To the right is an example of placement resolution. On move two both players move to E7, the resulting stone resolves to black by our rules. Note that the stone placed on move 4 does not resolve to white, as it is trapping.



2.3. **Capture rule of *SGo*.** After both White and Black move on current turn numbered n, using the stone placement resolution rule above we get some board state S.

Definition 1. *We say that a unicolor component C is* **trapped on turn** *n if C is trapped and some stone adjacent to C, or in C was placed on turn n. A stone is called* **trapped** *on turn n, if it is an element of C as above. We say that a red stone is trapped by white respectively black on turn n, if replacing it by a black respectively white stone it is trapped on turn n.*

A white component C is resolved as ***captured*** on this turn if the following is satisfied:

(1) C is trapped on this turn.
(2) If C is adjacent to a red stone that is adjacent to another trapped component then that component is white.
(3) One of the following holds:
  (a) No black or red stone adjacent to C is trapped by white on this turn.
  (b) (Suicidal capture) No stone in C is created on this turn. If a stone in C is adjacent to a stone trapped on this turn, then the latter stone is created on this turn and is not red.

In the case C is captured.

- The stones of C are removed as prisoners of Black as in Go.
- Any red stones adjacent to C resolve as black. (Concretely, they are replaced by black stones.)

If C is a black component the process is naturally mirrored. The above capture rule is invoked recursively until no captured on this turn n, unicolor components remain. At the end of this process, we further simplify the position as follows [2]. If any connected, trapped component C of red stones is adjacent only to white stones, and non of these white stones are trapped then all stones of C are replaced

[2] It is not logically necessary, unless we want a clean mathematical relationship of *SGo* with *Sym*(*Go*) outlined in Section 4.2.

by white stones. We call the whole process above, stage one reduction and the resulting board state will be denoted by $R_n(S)$.

Lemma 1. *The stage one reduction is well defined, that is independent of the order in which components are captured.*

Proof. Suppose we have two unicolor components C_0, C_1 captured on the given turn. By assumption if a red stone is adjacent to both C_0 and C_1 then C_0 and C_1 have the same color. But in that case it clearly does not matter in which order they are captured.

Thus suppose that no red stone is adjacent to both C_0 and C_1. If C_0 and C_1 have the same color they cannot be adjacent, that is there is no pair of stones, one in C_0 the other in C_1 that are adjacent. We now show that if they have opposite color then they are also not adjacent. Given this, if we apply capture rule to the group C_0 and then to the group C_1, this results in the same board state as first applying the capture rule to the group C_1 and then the group C_0.

We have the following possibilities. If C_0, C_1 both satisfy condition 3a, then they are trivially not adjacent. The same happens if C_0 satisfies the former condition and C_1 satisfies condition 3b. Suppose lastly that C_0, C_1 both satisfy condition 3b. And suppose by contradiction that they are adjacent. Then C_1 is adjacent to a stone in C_0, which by assumptions is trapped on this turn. Condition 3b applied to C_1 implies that the latter stone is created on this turn. But by assumptions no stone of C_0 is created on this turn. So we have a contradiction, and so C_0, C_1 are not adjacent.

To prove the general case proceed by induction. Let $S(n)$ be the sentence: for a collection $\{C_0, \ldots, C_n\}$ of captured unicolor components, the result of stage one reduction does not depend on the order of C_i. $S(1)$ was shown to hold. We show $S(n) \implies S(n+1)$. Let $\{C_0, \ldots, C_{n+1}\}$ be a given ordered collection of captured unicolor components, and let ρ be a permutation of the set $\{0, \ldots, n+1\}$. By the argument above and using a sequence of adjacent transpositions, the stage 1 reduction applied to $\{C_{\rho(0)}, \ldots, C_{\rho(n+1)}\}$ (respecting the given order) is the stage 1 reduction applied to $\{C_0, C_{\rho'(1)}, \ldots, C_{\rho'(n+1)}\}$ for some permutation ρ' fixing 0. Using the induction hypothesis on $\{C_{\rho'(1)}, \ldots, C_{\rho'(n+1)}\}$ we get that the latter stage 1 reduction is the stage 1 reduction applied to the order collection $\{C_0, \ldots, C_{n+1}\}$. $\square$

If there are no unicolor components trapped on the turn $n-1$ or lower, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction denoted by $R^k(S)$ satisfies:

$$R^1(S) = R_n(S).$$
$$R^k(S) = R_{n-k+1}(R^{k-1}(S)).$$

If there are no unicolor components trapped on the turn $n-k$ or lower, the capture stage is resolved after stage k reduction.

Remark 2. *This may appear to be fairly complicated, but in practice the resolution of most turns is easy to work out mentally, we will give some examples of such "nested entanglement states" further ahead. Of course having a computer work out turn resolution automatically is very helpful.*

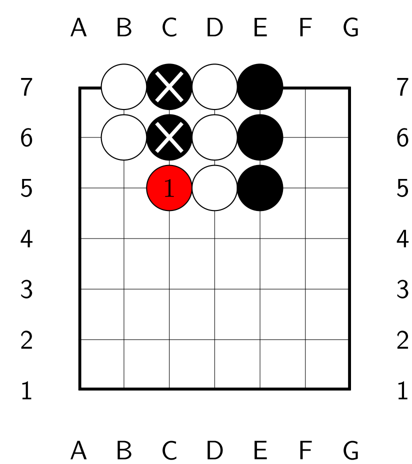
2.4. **End of the game.** *SGo* game ends after both players agree to end, or automatically after one of the players is mathematically winning, the latter can be determined by prisoner counts and or by geometry of the position. [3] At this point the winner is decided as in Go by territory and prisoner count. White's territory boundary consists of connected components of white/red stones. Blacks territory boundary consists of connected components of black/red stones. Red stones are never counted as prisoners. See Section 3.4 for an example.

A draw is now possible although compared to chess it must be much less likely, (on a typical 13 ×13 or 19 × 19 go board).

3. Examples

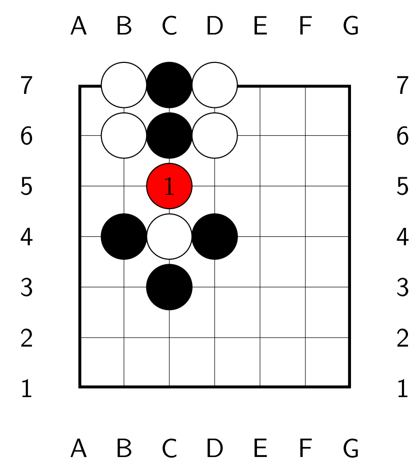
3.1. **Capture and non-capture moves.** Here are some examples and non-examples of capture.

On move 1 both Black and White move to C5. As this is a capture more, C5 will resolve to white at the end of the turn.

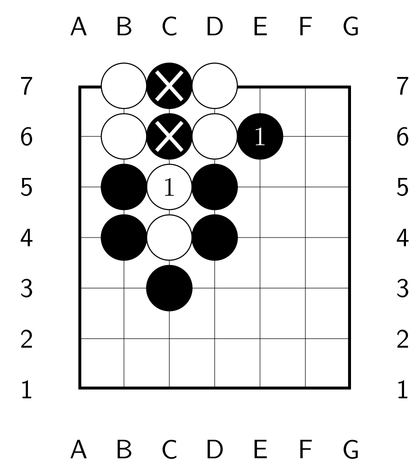


[3]In classical Go, Chess and likewise in SGo unless one puts hard limits on number of game moves, players may conspire to extend a game indefinitely and in fact infinitely unless there is a repetition clause (for Chess this is standard).

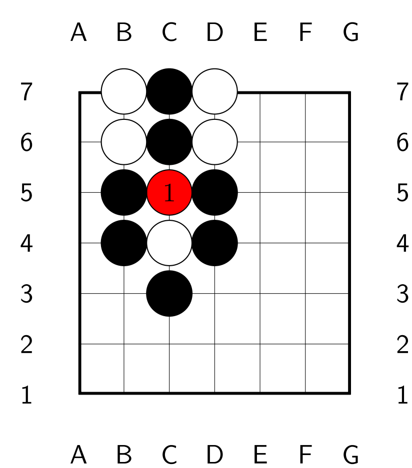
Not a capture move, as the created red stone is adjacent to an opposite color trapped component. This should make intuitive sense since any capture would be ambiguous from classical Go perspective.



White's move is a capture move, as Condition 3b is satisfied. This is consistent with classical Go behavior that suicidal captures resolve to capture.

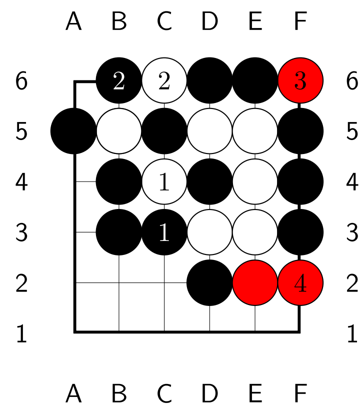


Not a capture move, now Condition 3b is not satisfied, since the stone placed on this turn is red.

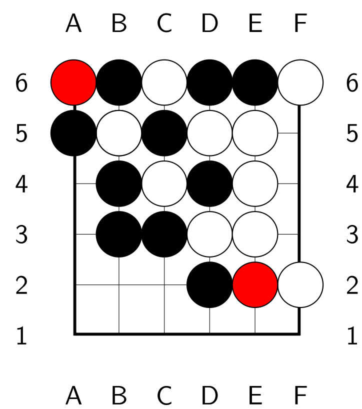


3.2. **More complex examples.** In the first example, White need only play with a pure strategy (and still be ostensibly winning). In more marginal, complex positions, the players may need mixed strategies.

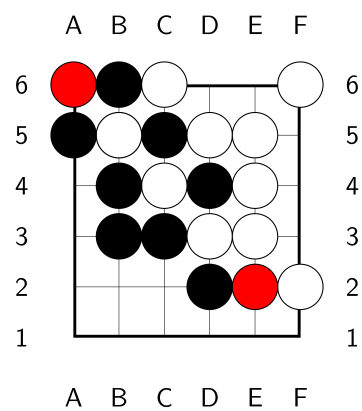
In the position on the right, on turn 1, White attempts capture at C4, while Black attempts capture at C3. Nothing is captured on turn 1 by our rules, and after turn 2 the position becomes an example of a "nested entangled state". A possible continuation is indicated. Note that F6 is not a capture by our rules, similarly to the last example of the previous section. But F2 is a capture move since condition 3a is satisfied.



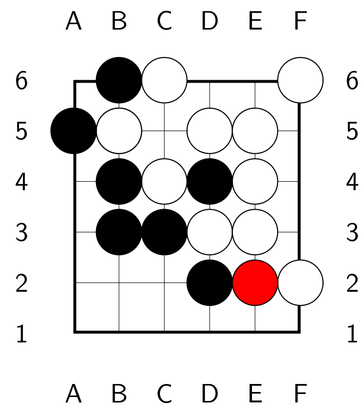
At the end of turn 4, stage one resolution is depicted on the right.



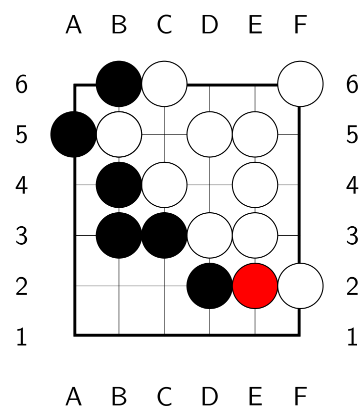
Stage two resolution.



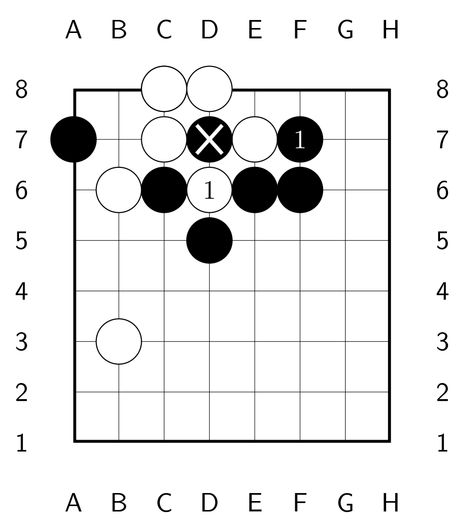
Stage three resolution.



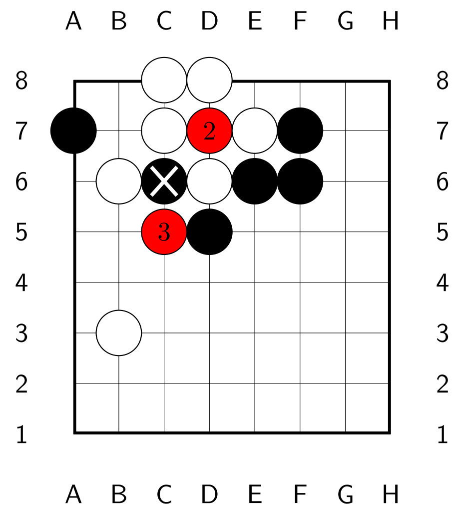
Stage four and final resolution of turn 4.



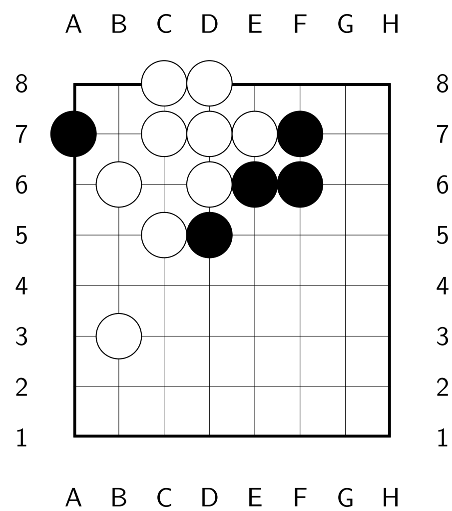
Ko type situation. Recall, that we don't have a Ko rule, but it is now unnecessary, since the board state cannot immediately loop.



Here is a possible continuation. Move 5 is a capture move, the stones surrounding B6 will resolve to black at the end of turn 5. The top white group cannot be saved.

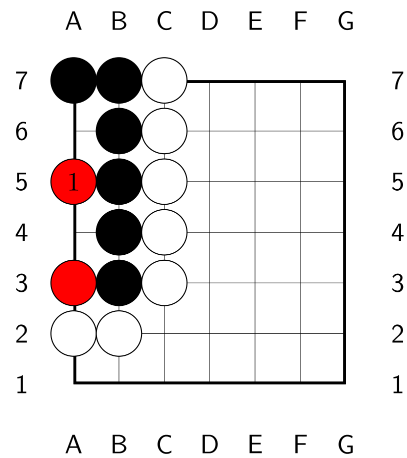


At the end of turn 3 we get the following position. The red stone at D7 is converted to white by the stage one reduction rule. To remark, this should make sense from our perspective of *SGo* as a "quantum analogue" of Go. A red stone at D7 is meant to be both black and white, but if it was a black stone we would have an illegal Go position, so it must actually be white.

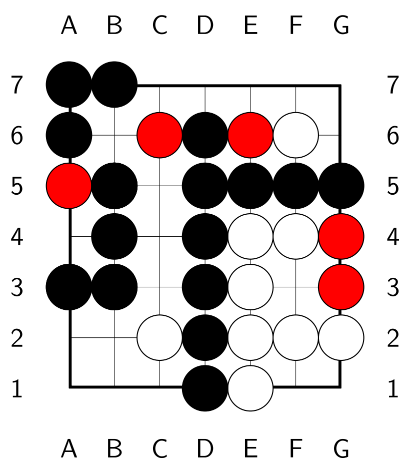


3.3. **Life and death.** Life and death is positionally very similar to Go, here is an example.

The black group is alive, that is impossible to kill.



3.4. **End game position.** In the diagram below, the game has ended. Assuming there are no prisoners, White has 3 points and Black has 19 points.



4. SGo as a symmetric simultaneous game

We will use the formalism of automata, as it is very flexible. The goal is to briefly introduce stateful, symmetric, simultaneous, games.

Definition 2. *A (stateful)* **simultaneous game** *G with 2 players P_0, P_1 or Black, White, consists of:*

- *A deterministic automaton G, whose set of states is a set $S(G)$ called the set of game states of G.*
- *The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \sqcup M_1(G)$: interpreted as possible moves of Black and White.*

- *Games states decompose as*
$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$
where $B(G)$ is some set (for example it might be the set of board states).
- *Nonempty subsets $W_0(G), W_1(G), D(G) \subset B(G)$ of final (accepting) states with empty pairwise intersection, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.*
- *A distinguished initial state $q_0 \in B(G)$, understood as the state in which the game is initialized.*
- *A partial game evolution function:*
$$E = E_G : S(G) \times M(G) \to S(G),$$
which has the properties:
  1. *$E(s, m)$ is undefined for s in one of the accepting states.*
  2. *If $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$ s.t. $E(s, m)$ is defined.*
  3. *If $(s, m) \in B(G) \times M(G)$, then $E((s, m), m') \in B(G)$ if defined, and is undefined if the grading of m is the grading of m'.*
  4. *Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined then $E(s, m_0, m_1)$ is defined.*
  5. *For s, m_0, m_1 as in the previous property, if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and*
$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

We recursively define:
$$E(s, m_1, \ldots, m_k) = E(E(s, m_1, \ldots, m_{k-1}), m_k).$$
We say that G is **finite** *if $S(G), M(G)$ are finite, and all game sequences have bounded length, where a game sequence is a sequence $\{m_i\}_{i=1}^{i=n}$ of moves s.t. $E(q_0, m_1, \ldots, m_k)$ is defined, for all $1 \leq k \leq n$.*

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 3. *We say that a simultaneous game G is* **symmetric** *if the following conditions are satisfied.*

1. *There is a bijection $\mathcal{S} : M(G) \to M(G)$ taking $M_0(G)$ onto $M_1(G)$.*
2. *There is a bijection $R : B(G) \to B(G)$ s.t. for all $s \in B(G)$, all $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, and for $\mathcal{S}$ as above we have:*
   - *$E(R(s), \mathcal{S}(m_0), \mathcal{S}(m_1)) = R(E(s, m_0, m_1))$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.*
   - *R also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.*

Lemma 3. *SGo is a symmetric simultaneous game.*

Proof. Set $G = SGo$, with latter described in Section 2. The set $M_0(G)$ corresponds to possible moves of Black and $M_1(G)$ the moves of White, as in the game description.

The state space is:
$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$

where an element of $B(G)$ is a legal board position and its entire legal game history. By legal, we mean the history is obeying the rules of Section 2.

The evolution map E is then obviously defined. For example if $s \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are legal moves then $E((s, m_0), m_1) \in B(G)$ is defined to be the resolution of the board position after applying the rules of Section 2, once Black's move m_0, and White's move m_1 are reflected on the board.

To see that G is symmetric, let $R : B(G) \to B(G)$ be the map that interchanges the black and white colors of the stones of the board states, leaving red stones invariant. And let $\mathcal{S} : M(G) \to M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \to M_1(G)$, which then induces $\mathcal{S}$. Then clearly the conditions of Definition 3 will be satisfied. □

4.1. **Solvability.** A theorem of Zermelo [3] says that for every suitably finite sequential game either Black or White has a winning strategy or there is a draw strategy. Obviously *SGo* cannot have a winning strategy, furthermore unless we put an upper bound on the length of a game, or on the number of position repetitions an *SGo* game length can be infinite. [4] So we will assume there is such a bound, which will make *SGo* a finite game.

Question 1. *Does SGo have a strategy to draw?*

Of course we have simultaneous games with no strategy to draw, the classical example is rock-paper-scissors. But the latter has a strategy to draw on average. The latter being a very trivial case of the existence of the Von-Neumann, Nash equilibrium for "matrix payoff games", [1]. *SGo* is not a matrix payoff game, but Nash's proof of the equilibrium theorem readily extends to this case. In general:

Theorem 1. *Assigning any payoff for a win and draw, a finite simultaneous game G in the sense above has a Nash equilibrium.*

Proof. For $s \in B(G)$ set

$$M_i(s) = \{m \in M_i(G) \,|\, E(s, m) \text{ is defined}\}.$$

The space of strategies of player P_i is then:

$$\mathcal{P}_i = \prod_{s \in B(G)} \Delta(M_i(s)),$$

where $\Delta(M_i(s))$ is the space of probability distributions over the finite set $M_i(s)$. Since $B(G)$ is also finite, the joint strategy space $\mathcal{P}_0 \times \mathcal{P}_1$ is a compact convex subspace of $\mathbb{R}^N$, for some very large N.

The expected payoff function on $\mathcal{P}_0 \times \mathcal{P}_1$ is easily seen to be continuous, due to finiteness of G. [5] Using this, we get that the multi valued function ϕ, assigning to a joint strategy its set of countering strategies, has a closed graph. Furthermore, the image by ϕ of any point is non-empty and convex.

Then as argued by Nash [1] using Kakutani's fixed point theorem we get existence of an equilibrium point. □

Corollary 1. *A finite symmetric simultaneous game G has a strategy to draw on average, and in particular SGo has such a strategy.*

[4] This is also true for Go and Chess.

[5] Possibly finiteness condition can be relaxed.

Proof. Assign 1 for a win, -1 for a loss and 0 to a draw. Then it is enough to show that for any Nash equilibrium strategy the expected payoff is 0 for both players.

Suppose otherwise, and suppose without loss of generality that

$$(\mathcal{D}_0, \mathcal{D}_1) \in \mathcal{P}_0 \times \mathcal{P}_1$$

is an equilibrium joint strategy where Black has a positive expected payoff.

Let $R, \mathcal{S}$ be the symmetry operators as in the proof of Lemma 3. Naturally applying these operators to the joint strategy $(\mathcal{D}_0, \mathcal{D}_1)$ we get that there is an equilibrium joint strategy $(\mathcal{D}'_0, \mathcal{D}'_1)$ where White has positive expected payoff. But this is absurd. □

4.2. **SGo as a symmmetrization of Go.** For any sequential game G there is a canonically associated symmetric simultaneous game $Sym(G)$, which we may call its universal symmetrization. It is not in our scope to discuss this in detail, and is likely well understood by experts. A state in this game is a set of game histories in G, this should not be surprising as states of *SGo* also can be clearly interpreted as some sets of legal game histories in *Go*. (This is only strictly speaking true if we impose the optional rule in stage one reduction, of Section 2.3).

From one perspective *SGo* is $Sym(Go)$ with a history reduction mechanism, which is built into the rules of *SGo*. The latter gives *SGo* dynamic behavior similar to *Go*. It is interesting to wonder if there is something similar for chess.

References

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[2] R. Penrose, *Emperor's new mind*, 1989.

[3] U. Schwalbe and P. Walker, *Zermelo and the early history of game theory*, http://www.math.harvard.edu/~elkies/FS23j.03/zermelo.pdf.